

Nagaraj Vernekar

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Research interest

Galactic Archaeology
Asteroseismology

Chemical evolution
Massive stars

Stellar spectroscopy
Binary systems

Education

Oct 2022 – Sep 2025

Ph.D. in Astrophysics

Università di Padova

Topic : Galactic Archeology with Machine learning and novel data

Supervisor : Dr. Sara Lucatello and Dr. Lorenzo Spina

Sep 2019 – Jan 2022

Master in Astronomy and Astrophysics

Katholieke Universiteit Leuven

Topic : On the photometric and spectroscopic variability of Be stars: the case of HD 93683

Supervisors : Dr. Dominic M. Bowman and Dr. Julia Bodensteiner

Experience

Jan 2026 – present

Postdoctoral researcher

University of Oklahoma

Supervisor : Prof. Michael Hayden

Feb 2022 – Sept 2022

Research Assistant

Topic : Blue straggler stars in globular clusters

Indian Institute of Astrophysics

Supervisor : Prof. Annapurni Subramaniam

Publications

First Author

- Vernekar, N., Hayden, M. and Lucatello, S., 2026, "Stellar parameters and abundances for 3.2 million DESI DR1 spectra using machine learning", Submitted to ApJS
- Vernekar, N., Lucatello, S., Spina, L. et al., 2026, "LRPayne: Stellar parameters and abundances from low-resolution spectra", *A&A*, 706, A217
- Vernekar, N., Lucatello, S., Kuzma, P., Spina, L., 2025, "Kinematics of metallicity populations in Omega Centauri using Gaia Focused Product Release and Hubble Space Telescope", *A&A*, 696, A168
- Vernekar, N., Lucatello, S., Bragaglia, A. et al., 2024, "Stellar population astrophysics (SPA) with the TNG. Abundance analysis of nearby red giants and red clump stars: combining high resolution spectroscopy and asteroseismology", *A&A*, 684, A85.
- Vernekar, N., Subramaniam, A., Jadhav, V. and Bowman, D., 2023, "Photometric variability of blue straggler stars in M67 with TESS and K2", *MNRAS*, 524, 1360-1373.

Publications (continued)

- Other** ■ M. Dal Ponte, V. D'Orazi, A. Bragaglia, A.R. Casey, N. Storm, L. Spina, J. Alonso-Santiago, G. Andreuzzi, A. Frasca, J. Kos, S. Lucatello, D. Romano, A. Vallenari, **Vernekar, N.**, 2025, "Stellar Population Astrophysics (SPA) with the TNG. Non-local thermodynamic equilibrium atmospheric parameters and abundances of giant stars in 33 open clusters", *A&A*, 701, A289
- Panthi, A. and Vaidya, K., **Vernekar, N.**, Subramaniam, A., Jadhav, V. and Agarwal, M., 2023, "UOCS - XI. Study of blue straggler stars in open cluster NGC 7142 using UVIT/AstroSat", *MNRAS*, 527, 8325-8336.

Observing proposal

- Jan 2026 - Present ■ **PI** of 4 proposals for ARCES at 3.5m Telescope in Apache Point (50 hrs)
- Jan 2026 ■ **Co-PI** of supplementary proposal for 4MOST survey
- Jan 2025 ■ **PI** of proposal for GIARPS (HARPS-N and GIANO) at TNG (7.15 hrs)
- April 2024 ■ **Co-PI** of proposal for PEPSI at LBTO (2.18 hrs)
- July 2023 ■ **Co-PI** of proposal for PEPSI at LBTO (3.41 hrs)

Conference and seminars

- Mar 2026 ■ **4MOST SV hackathon**, Heidelberg, Germany
- Jul 2024 ■ **EAS 2024** (Contributed talk), Padua, Italy.
Title: Omega Cen - through the FPR eye
- May 2024 ■ **The Milky Way Assembly Tale** (Poster), Bologna, Italy.
Title: Kinematics of metallicity populations within Omega Cen using Gaia FPR
- Mar 2024 ■ **SPA-OC Workshop** (Contributed talk), Bologna, Italy.
Title: SPA - not only open clusters
- Oct 2023 ■ **STARS Across the Universe** (Contributed talk), Naples, Italy.
- Jul 2021 ■ **EAS 2021** (Poster + flash talk), held virtually.

Grant

- Oct 2022 – Sep 2025 ■ **CARIPARO Scholarship** (65,000 euros)
- Oct 2020 – Sep 2021 ■ **Devaraj Urs Foregin Scholarship** (20,000 euros)

PhD schools and workshops

- Feb 2025 ■ **NEON Observational School**, Calar Alto, Spain
- Aug 2024 ■ **Porto Summer School on Asteroseismology**, Porto, Portugal
- Aug 2023 ■ **MESA summer school**, Budapest, Hungary
- June 2023 ■ **PHOEBE workshop**, Ljubljana, Slovenia
- Feb 2023 ■ **ORP Proposal Writing School**, virtually hosted

Public Outreach

- Apr 2022 – Mar 2023  **Project Mentorship**, Naxxatra Sciences and Collaborative Research
Mentored over 90 bachelor students on different astrophysics projects
- Aug 2022  **Workshop Lead**, Observational Astrophysics, Indian Institute of Science, Bengaluru
About 70 students were taught different aspects of astronomy using hands-on approach
- Jul 2022  **Naxxatra Astrophysics seminar** (Invited talk), Bengaluru
Title: Essential techniques in Spectroscopy
- April 2022  **Naxxatra Astrophysics seminar** (Invited talk), Bengaluru
Title: Observational techniques in Astronomy

References

Dr. Michael Hayden
Professor,
University of Oklahoma,
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Dr. Lorenzo Spina
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